Conditional Generation of Lithium-Vacancy Configurations for Cathode Relithiation

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Abstract

We study conditional generation of lithium-vacancy patterns in $LiCoO_2$, a battery cathode material. Each example specifies which lithium sites are occupied, and the generator is asked for new patterns at a given lithium content and energy range. We compare conditional variational autoencoders (CVAEs), autoregressive models and simple sampling baselines, measuring distinct qualifying configurations per attempt. With energies evaluated by the pretrained neural potential UMA, the spatial CVAE, its autoregressive control and mutation achieve 62.20%, 63.04% and 58.37% balanced yield. Neighbor information improves yield, whereas increasing latent dimension improves reconstruction without a consistent sampling benefit. Controlled interventions show that the energy condition influences generation much more than the latent vector. Training on energies after atomic relaxation also fails to improve the matched relaxed-energy test. These comparisons support spatial conditional sampling without establishing a latent-specific advantage, and show why useful latent learning needs validation beyond reconstruction.

1 Research question, motivation and related work

A lithium-ion battery cathode stores and releases lithium during cycling. Recycling can restore this material by replenishing missing lithium rather than breaking it down into its constituent elements. Hydrothermal relithiation, which uses heated water-based processing, has regenerated spent lithium cobalt oxide, $LiCoO_2$ (LCO) [1]. At a fixed lithium content, there are many ways to distribute the occupied and empty lithium sites. These arrangements can have different energies because their neighboring atomic environments differ [2]. Finding useful arrangements is therefore a conditional generation problem with a combinatorially large discrete output space.

Energy is a property of the atom types, positions and repeating simulation cell. Density functional theory (DFT) calculates an approximate electronic structure and serves as a reference for these energies and the forces on atoms, but is expensive for large datasets. A machine-learned interatomic potential (MLIP) is a model trained to approximate such calculations. It predicts energy from an atomic structure; forces describe how that energy changes when atoms move. Here, pretrained MLIPs provide inexpensive energy labels and score generated samples. A *single-point* evaluation leaves all positions fixed, whereas *geometry relaxation* repeatedly moves atoms using the predicted forces to approach a local energy minimum. Generating a vacancy pattern alone does not perform that minimization.

We ask: **can a CVAE generate new configurations in a requested energy range more efficiently than uniform sampling and local mutation, and which representations improve this yield?**

Wakamiya and Ishikawa [3] study a related problem for platinum/nickel catalyst surfaces. Their convolutional CVAE learns grids specifying which element occupies each site. It conditions on favorable classes of formation energy (a stability measure) and oxygen-reduction overpotential (the extra voltage needed to drive a catalytic reaction), then iteratively evaluates and retrains on generated

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arrangements. Their evaluator already uses UMA, a universal MLIP [4]. Our contribution is a controlled study of periodic vacancy patterns: exact composition, symmetry-aware novelty, neighbor representations, and a model without latent variables distinguish the causes of any sampling gain.

Targeting several energy ranges, including low-energy configurations, is useful for assembling MLIP fine-tuning data. The potential energy surface maps atomic arrangements to energy; covering diverse arrangements at different energies can provide information beyond repeatedly sampling a few minima. Our energy classes are intended to support that coverage. Whether the resulting data actually improve a fine-tuned potential remains a separate validation question.

2 Training data and structural representation

What a configuration contains. We start from a repeating cell with 48 lithium, 48 cobalt and 96 oxygen atoms, $Li_{48}Co_{48}O_{96}$. Its atomic arrangement is called *O3*: one particular layered structure in which lithium layers alternate with cobalt–oxygen slabs. Periodic boundaries repeat the cell in all three directions, so a site near one cell boundary can neighbor a site across the opposite boundary. Removing selected lithium atoms creates a configuration. The 144 cobalt/oxygen atoms, the remaining lithium coordinates and the cell stay fixed for the main dataset; these examples are *not* geometry-optimized vacancy structures.

An occupation mask $o \in \{0,1\}^{48}$ records $o_i = 1$ for a present lithium atom at site i and $o_i = 0$ for an empty site, or vacancy. The count $n = \sum_i o_i \in \{12, 24, 36\}$ corresponds to lithium fractions $x = n/48 = 0.25, 0.50, 0.75$; the structure contains $144 + n$ atoms. UMA-S-1.2.1 with the `omat` setting supplies training energies. MACE-MH-1 with `omat_pbe` independently scores the same generated structures [4, 5]. These are two pretrained energy models, not two independent DFT reference datasets.

Periodicity and duplicate structures. An allowed translation, rotation or reflection of the periodic parent can turn one mask into another without changing its physical arrangement. All masks related by the 192 retained parent-preserving site permutations form a *symmetry orbit*. We choose one canonical representative per orbit and group equivalent masks before splitting the data. This prevents rotated or translated copies from appearing as independent training and validation examples. The split contains 40,157 training, 4,967 validation and 5,123 test orbits. Only training examples set model weights and energy thresholds; validation selects checkpoints. Test examples are excluded from these decisions. For generation assessment, novelty also excludes a fixed reference of 344,001 previously observed orbits, including the training data.

Why neighboring environments matter. A binary vector tells the model which sites are occupied but does not explicitly say which sites are physically close. Two masks with the same count can describe clustered or dispersed vacancies. We therefore compare the ordered vector with graph-based encoding, and give the decoder summaries of neighboring site states. These are *features*: numerical descriptions that make physically relevant relationships accessible to the network. All features respect periodic neighbors. The examples explore lithium ordering within one fixed O3 framework; other layer stackings such as O2, three-dimensional spinel frameworks and atomic rearrangements are absent. Especially at low lithium content, the starting framework need not be stable. The model consequently learns a restricted configuration distribution, not which crystal structure is chemically preferred.

3 Conditional generation and evaluation

What the condition requests. Energy labels are fitted separately at each lithium count. Training quantiles q_{20} and q_{80} divide energy per present atom, e in eV/atom, into LOW ($e \le q_{20}$), MIDDLE ($q_{20} < e \le q_{80}$) and HIGH ($e > q_{80}$), with equal-energy ties kept together. The four-component condition $c = (n/48, \mathrm{onehot}(b))$ combines lithium fraction with the requested class b; for example, LOW has the three class entries $(1, 0, 0)$. A low-energy request asks for a learned range, not an optimization gradient or a guarantee of the lowest possible energy. Separate thresholds prevent the composition-dependent energy baseline from determining the class. Each MLIP uses its own training-derived thresholds when evaluating proposals.

How a CVAE generates a mask. The encoder sees the complete training mask and condition. It returns a Gaussian $q_\phi(z \mid o, c) = \mathcal{N}(\mu_\phi, \operatorname{diag} \sigma_\phi^2)$ over a short latent vector $z \in \mathbb{R}^D$. The mean μ_ϕ and variance σ_ϕ^2 are learned functions of the input. During training, $z = \mu_\phi + \sigma_\phi \odot \epsilon$, with $\epsilon \sim \mathcal{N}(0, I)$, allows gradients through the random sample. During generation there is no input mask: z is drawn from the fixed prior $p(z) = \mathcal{N}(0, I)$, and the decoder constructs a new mask [6, 7].

A gated recurrent unit (GRU) with 64 hidden channels makes the 48 occupied/vacant decisions in a fixed order. The probability factorization and training loss are

$$p_\theta(o \mid z, c) = \prod_{i=1}^{48} p_\theta(o_i \mid o_{<i}, z, c, h_i), \qquad \mathcal{L} = -\mathbb{E}_{q_\phi} \log p_\theta(o \mid z, c) + \beta_t D_{\mathrm{KL}}(q_\phi \| p). \quad (1)$$

Here θ, ϕ are decoder/encoder parameters, $o_{<i}$ contains the decisions before site i, and h_i summarizes their neighboring environments. The first loss term rewards reconstruction of training masks. The Kullback–Leibler (KL) term encourages the encoder distributions to resemble the prior from which we will sample; β_t controls its weight at update t. Both terms use natural logarithms and are measured in nats per mask. Training supplies the true preceding bits (teacher forcing); generation uses preceding sampled bits. To enforce the requested count exactly, later decisions are forced vacant when no lithium remains, or occupied when every remaining site is needed.

What the model comparisons isolate. The dense encoder uses 128 then 64 hidden units to process the ordered mask and condition. The graph encoder instead represents atoms and candidate lithium sites as nodes, retaining empty lithium sites so their locations are known. Two width-32 message-passing layers exchange information over periodic neighbors within 5 Å, using six distance channels; pooled host and lithium-site features produce the Gaussian parameters [8].

The decoder has a different information constraint. Before deciding a site, it can know which nearby sites are already occupied, already vacant, or not yet assigned, but cannot see future target bits. The six-feature baseline summarizes these three states using nearest-neighbor and exponentially distance-weighted neighborhoods. The nine-feature alternative separates them into three distance shells at 2.811, 4.869 and 4.912 Å, including neighbors in adjacent layers. This directly supplies local geometry while preserving causal generation.

Table 1: Model comparisons. All listed CVAEs have $D = 4$. AR denotes autoregressive sampling without an encoder or latent vector.

Model	Encoder	Question tested
A: graph CVAE	Graph Gaussian	Do latent variables help when spatial context is supplied?
B: spatial AR	None	Can the same spatial decoding rule work without a latent?
C: dense CVAE	Dense Gaussian	Does a different encoder improve generation with the same decoder?
D: scrambled AR	None	What happens when spatial associations are scrambled?
A-shell/B-shell	Graph/none	Does resolving three neighbor shells improve yield?

The AR control retains the decoder width, condition and neighbor features while removing z. It still samples each decision randomly. The scrambled control reassigns neighbors nonphysically while retaining overall weights and parameter count; this also changes when information becomes available along the sequence, so it does not isolate distance alone. Shell features add 576 GRU parameters, and graph/dense encoders have different capacities. These comparisons are informative but not all strictly capacity-matched. Fixed-order decoding also means the full model is not permutation-equivariant; canonicalization handles the retained parent symmetries.

Training and model selection. Each fit uses Adam, learning rate 10^{-3}, batch size 512 and 4,000 updates. The KL weight gradually rises to 0.1. Training class weights compensate for the roughly 20/60/20 LOW/MIDDLE/HIGH imbalance; validation keeps the original example weights. Checkpoints minimize validation reconstruction plus 0.1KL, using fixed random noise for comparable validation estimates. Model selection uses development comparisons; the final sampling assessment does not select or modify models.

What counts as successful generation. Uniform sampling chooses exactly n lithium sites at random. Mutation starts from a training mask in the requested energy class and swaps one, two or

three occupied sites with the same number of vacancies, preserving composition. These baselines test whether learning improves on random exploration and on small edits to already useful examples.

Our primary score asks how many attempts produce a genuinely new arrangement in the requested range. For replicate s, count n and class b,

$$Y_{snb} = \frac{\#\{\text{distinct valid orbits absent from the reference and measured in class } b\}}{\#\{\text{all attempted proposals at } (s, n, b)\}}. \tag{2}$$

Generating one qualifying arrangement ten times earns one discovery, not ten; repeats, invalid outputs and failed requests still consume attempts. We average the nine count/class cells equally to obtain *balanced yield*, then report the mean and standard deviation (SD) across fitted-model or baseline-sampling replicates. Each cell receives 128 attempts. The comparison includes 44,928 primary attempts and 4,992 separate latent probes. Duplicate physical queries are evaluated only once, without removing their attempts from the denominator.

4 Representation and hyperparameter results

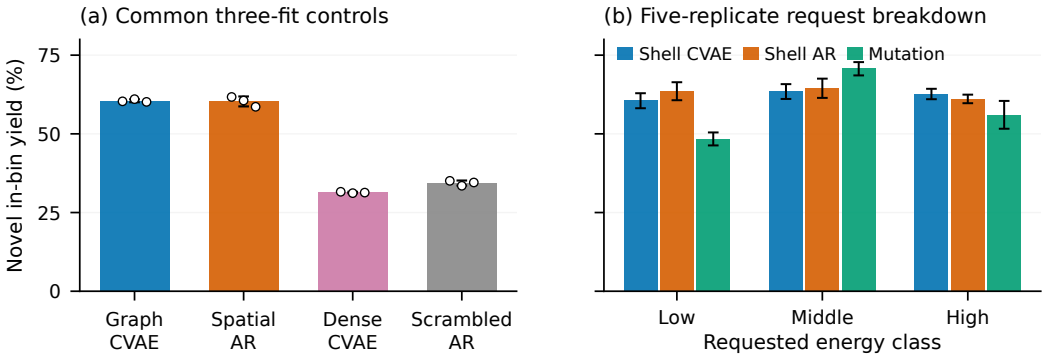


Figure 1: How representation changes useful sampling. Left: three matching fits per control. Right: five fit/sampling replicates per energy request. Bars show means, dots individual fits, and whiskers replicate SD. All energies use UMA. Mutation is stronger for MIDDLE requests, so gains depend on the requested energy range.

Neighbor features help both the CVAE and AR. With the same six decoder features, the graph CVAE and spatial AR have similar yield: their paired difference over three matching fits is $+0.20 \pm 1.49$ percentage points (pp). Dense encoding and scrambled spatial associations give only $31.37 \pm 0.22\%$ and $34.38 \pm 0.80\%$ (Figure 1). Thus an encoder that can process the complete mask is not sufficient; how the decoder receives neighboring information matters.

Separating three distance shells improves balanced UMA yield by 1.96 ± 0.32 pp for the CVAE and 2.83 ± 1.46 pp for AR over five paired fits. Table 2 shows that both outperform uniform sampling and mutation in balanced yield, with the shell AR giving the highest point estimate under both teachers. The CVAE does not outperform this matched AR control (-0.83 ± 1.13 pp under UMA). This does not prove that latent variables can never help; no advantage is demonstrated at this data and training budget. The gain over mutation is also not uniform: MIDDLE yield is 63.44% for the shell CVAE and 64.48% for shell AR, versus 70.68% for mutation.

More latent dimensions improve reconstruction, not reliably yield. A separate experiment varies $D \in \{4, 8, 16, 32\}$ in a dense-encoder CVAE with the GRU decoder but no spatial decoder features, with three fits per dimension and the same 4,000-update budget (Figure 2). Validation reconstruction loss improves from 17.01 to 14.87 nats as D grows, but useful yield ranges from 28.82% to 32.64% without a monotonic trend. It stays near uniform sampling and below mutation (59.72%) within that experiment. Four dimensions provide an economical default, but this is not evidence that four is optimal for the graph/shell architecture. The dimension experiment uses a separate development proposal pool, so its absolute yields should not be treated as a direct feature comparison with Table 2.

Table 2: Balanced novel in-request yield (%, mean ± SD), five fit or sampling replicates per row. MACE evaluates the same UMA-trained proposals using its own training-derived energy classes. Agreement is a second-model check, not DFT validation.

Method	UMA	MACE
Graph CVAE, six features	60.24 ± 0.53	61.72 ± 0.72
AR, six features	60.21 ± 1.23	62.03 ± 0.76
Graph CVAE, shell features	62.20 ± 0.56	64.03 ± 0.79
AR, shell features	63.04 ± 1.47	64.38 ± 0.96
Mutation	58.37 ± 1.08	57.97 ± 1.39
Uniform fixed-count	33.32 ± 0.04	33.32 ± 0.04

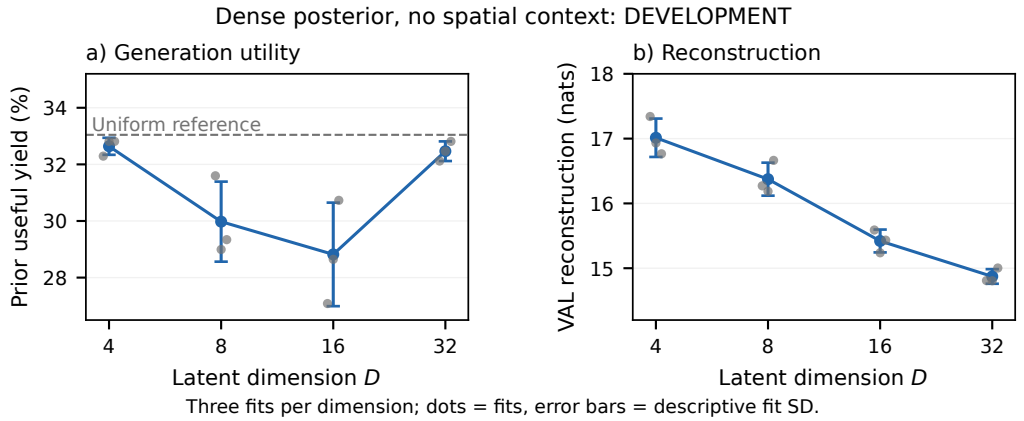


Figure 2: Latent-dimension sensitivity for a dense encoder and GRU decoder without spatial features. Three fits per dimension show better validation reconstruction with larger D, but no consistent improvement in prior-sampling yield. Error bars are fit SD. This is a finite sensitivity study, not demonstrated convergence.

Class weighting and checkpoint choice. A dense-encoder/GRU experiment at $D = 8$ asks whether rare energy classes need greater training weight, and whether the validation criterion overemphasizes prior matching. Table 3 compares equalized versus unadjusted training class weights and two KL coefficients used to *select checkpoints*. The training loss still ends at $\beta = 0.1$ in every case; changing checkpoint selection is different from changing the training objective. The combination of class balancing and selection coefficient 0.1 gives the best yield in this comparison, showing that the validation objective can affect downstream sampling even when the training objective is unchanged.

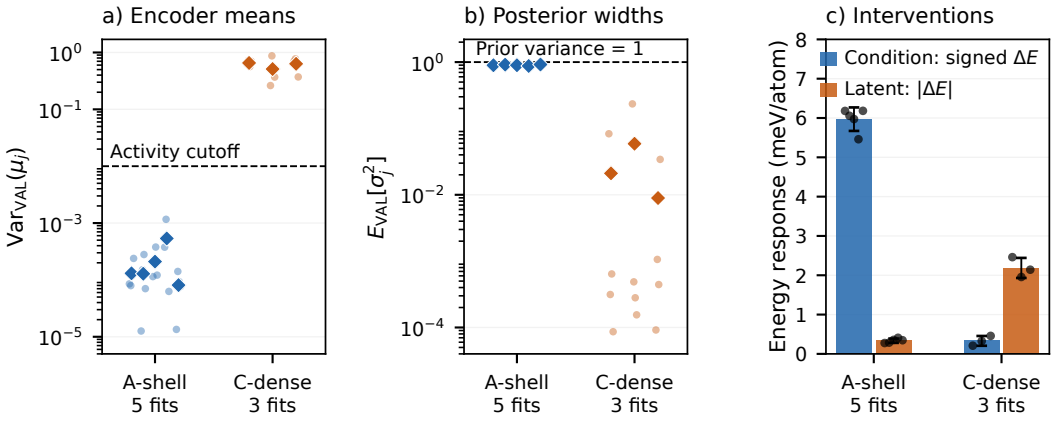
Table 3: Dense-encoder/GRU $D = 8$ development yield (%, mean ± SD over three fits). Columns change the validation selection criterion; rows change training weights.

Training class weights	Selection KL weight 1	Selection KL weight 0.1
Unadjusted	29.28 ± 1.91	30.44 ± 0.66
Class-balanced	30.61 ± 0.27	32.12 ± 0.63

A one-fit screen reducing the *training* KL weight to 0.03 raises balanced yield by 0.69 pp but lowers the separately tracked fixed-threshold low-energy yield by 1.04 pp. This is not consistent evidence of improvement. Adding explicit remaining-count features to spatial AR gives 61.40% in three fits, below the shell AR point estimate. Learning rate, batch size, widths, graph cutoff and training duration were held fixed; no convergence claim is made for them. The dimension study also changes parameter count with D, and further training could change the comparison. The present evidence supports a compact model with useful spatial context, not a globally optimized set of hyperparameters.

5 Understanding the learned latent space

The latent vector is intended to be a short code for an arrangement, but good reconstruction does not establish that this code helps generate useful new samples [9]. Three questions must be separated: does the encoder store information about a mask, does the decoder use z when sampling, and does the explicit energy request control the output? We examine all three.



Dots: components (a–b), fit means (c). Diamonds: fit means. Error bars: fit SD.

Figure 3: Encoder statistics and controlled sampling tests. A-shell uses five fits; C-dense uses three and the six-feature decoder, so this is not an encoder-only ablation. Components within a fit are not independent replicates. The 0.01 activity cutoff is heuristic. The last panel compares a signed LOW-to-HIGH energy shift with an absolute change from varying z; these are distinct observables. Diamonds and energy error bars show fit means and SD.

A wide distribution can still carry little information. For a useful encoder, different masks might produce distinguishable latent means or variances. In the shell CVAE, mean posterior variances are 0.854–0.943, close to the unit variance of the prior, while posterior means barely change across masks: their variances span 1.26×10^{-5}–1.16×10^{-3}. Using $\mathrm{Var}(\mu_j) > 0.01$ as a heuristic for an active coordinate gives zero active coordinates in all five fits. Lowering that cutoff to 0.001 makes one coordinate active in one fit. This supports near *posterior collapse*: the encoder distribution becomes almost independent of the input. It does not mean that all samples have the same z, or that the latent distribution has shrunk to one point; the distributions remain broad. A plausible mechanism is that the decoder can predict from the condition and partial pattern without needing z, while the KL term penalizes encoding extra information. This interpretation is consistent with the no-latent control.

How much information can the code contain? The average validation KL is only 0.0072–0.0202 nats per mask (mean 0.0123). To relate this to information, average encoder distributions over masks with the same condition to obtain $q(z \mid c)$. Under the weighted empirical validation distribution, the exact identity is

$$\underbrace{\mathbb{E}_{o,c} D_{\mathrm{KL}}(q_\phi(z \mid o,c) \| p(z))}_{\overline{K}} = \underbrace{I_q(o; z \mid c)}_{\text{information about the mask}} + \underbrace{\mathbb{E}_c D_{\mathrm{KL}}(q(z \mid c) \| p(z))}_{\text{averaged distribution's mismatch to the prior}} . \quad (3)$$

Adding and subtracting $\log q(z \mid c)$ inside the KL separates the two terms. Conditional mutual information $I_q(o; z \mid c)$ measures how much observing z reveals about o after the supplied condition is already known. Both right-hand terms are nonnegative, so $I_q(o; z \mid c) \le \overline{K}$. Dividing the largest observed KL by $\log 2$ gives an upper bound of approximately 0.030 bits per mask. This bounds information under the stated empirical distribution; it is not a direct mutual-information estimate, a population guarantee or a physical configurational entropy. A large KL, conversely, need not mean useful information: it can also reflect mismatch between the averaged encoder and the sampling prior.

Changing the condition has a much larger effect than changing the code. The CVAE's autoregressive decoder has two sources of randomness: the sampled latent z and the random numbers U used for its occupied/vacant decisions. We can write its output as $o = F_\theta(c, z, U)$. To isolate energy conditioning, we keep z, U fixed and change LOW to HIGH. The shell CVAE's mean UMA energy increases by 5.97 meV/atom, where 1 meV is 10^{-3} eV. To isolate latent influence, we instead keep c, U fixed and resample only z. The mean absolute energy change is just 0.335 meV/atom; on average 0.851 of the 48 occupation bits change, and 65.42% of pairs remain in the same symmetry orbit. The energy request clearly affects the output, while the latent contributes relatively little. Diversity can still come from U, which also lets a no-latent AR generate different masks.

The dense CVAE illustrates the converse. Its KL is 12.52–13.32 nats and all four coordinates are active. Changing z alters 13.04 occupation bits on average, yet changing LOW to HIGH shifts energy by only 0.331 meV/atom and its useful yield is poor. An active code therefore does not guarantee effective conditional sampling from the prior. The dimensions also have no assigned physical meaning; their signs and order can change across fits. These tests describe what the learned code does without assigning arbitrary axes to physical mechanisms.

6 Relaxation targets and limitations

Can the generator learn which masks become low-energy after atoms move? For a given mask, define E_{del} as the single-point energy at the fixed starting coordinates and E_{end} as the energy reached after fixed-cell atomic relaxation. The latter is a local endpoint, not a certified global minimum. A paired dataset of 6,000 masks assigns 3,600 training, 1,200 validation and 1,200 test roles before labeling. UMA relaxation uses FIRE, a force-based local optimizer, and stops when the largest atomic force is below 0.05 eV/Å, subject to 500-step and 120-second caps. Eight interruptions are retained; usable fit data contain 3,594 training and 1,200 validation pairs.

We train three graph/shell CVAEs per energy target using matching initializations, architecture and 4,000 updates. Only labels and training-derived class thresholds differ. This directly tests whether knowing relaxed energies improves the choice of masks; both models still output masks, and neither predicts relaxed coordinates. Generated samples must actually be relaxed before evaluating E_{end}.

At each lithium count, every generator is assessed against the same endpoint bins fitted on the paired training data. Across 10,944 attempted proposals, 7,616 distinct physical queries all converge. The six-feature graph CVAE, spatial AR and mutation reach endpoint LOW yields of 65.63%, 64.06% and 53.82%, respectively, in three replicates; CVAE minus AR is 1.56 ± 2.76 pp. In the paired-data comparison, however, training on E_{end} gives 60.42% endpoint LOW yield versus 62.50% when training on E_{del} (-2.08 ± 2.76 pp; no positive replicate). Balanced endpoint yields are 58.74% versus 59.14%. Relaxed labels therefore do not improve this controlled test. The large-corpus and paired-data models use different data and neighbor features, so their absolute yields are not an architecture comparison. Endpoint energies also require their own thresholds: relaxation lowers energies, so applying unrelaxed thresholds would inflate apparent success.

What these results establish. Spatial conditional sampling produces new configurations in requested energy ranges more efficiently than uniform masks, but a benefit specific to the variational latent pathway remains unestablished. Replicate SD describes fit variability, not a confidence interval; a few fits do not prove equivalence. Symmetry-aware novelty establishes a different arrangement, not a distinct relaxed minimum. The restriction to one O3 parent, three compositions and two MLIPs leaves transfer to other sizes, materials and crystal frameworks untested. The energy labels have not been established as DFT-accurate, and generated frequencies are not equilibrium probabilities. Vacancy sampling also omits atomic displacement, strain and magnetic variation, so energy-bin diversity covers only part of the potential energy surface relevant to MLIP fine-tuning.

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